

FILTER DESIGN TEST CASE

SAW (Surface Acoustic Wave) - WiFi 5 GHz for GPS/GNSS Anti-Jamming Filter
Design Validation and Performance Characterization

DOCUMENT INFORMATION

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1. OBJECTIVE

This document describes the design, simulation, and characterization of a SAW (Surface Acoustic Wave) filter targeting WiFi 5 GHz for GPS/GNSS Anti-Jamming Filter. The filter is designed on AIN on SiC substrate with a target center frequency of 5436.3 MHz and bandwidth of 220.8 MHz. The objective is to validate that the filter meets the required insertion loss, rejection, and linearity specifications for integration into the GPS/GNSS Anti-Jamming Filter module. This design iteration addresses feedback from the previous revision regarding out-of-band rejection improvements and temperature stability.

2. SCOPE

- Filter Type: SAW (Surface Acoustic Wave)
- Frequency Band: WiFi 5 GHz
- Application: GPS/GNSS Anti-Jamming Filter
- Substrate: AIN on SiC
- Package: CSP (Chip Scale Package)
- Design Tool: Matlab RF Toolbox R2024b
- Design Center: San Jose Advanced Design Lab

This specification applies to all variants of the SAW (Surface Acoustic Wave) design targeting WiFi 5 GHz. Results are used to qualify the design for mass production release.

3. DESIGN PARAMETERS

Center Frequency:	5436.3 MHz
Bandwidth (3 dB):	220.8 MHz
Insertion Loss Target:	≤ 3.4 dB
Return Loss Target:	≥ 18.0 dB
Out-of-Band Rejection:	≥ 46.0 dB
Isolation (Duplexer):	≥ 27.0 dB
Q Factor:	4117
Temperature Coefficient:	-10.0 ppm/°C
Die Size:	2.4 x 2.4 mm
Wafer Size:	6-inch
Number of Resonators:	12
Electrode Material:	Mo/Al

Passivation: SiO2

4. TEST PROCEDURE

1. Open Matlab RF Toolbox R2024b and load the SAW (Surface Acoustic Wave) template project.
2. Define the substrate stack: AlN on SiC with specified layer thicknesses.
3. Set design targets: center freq 5436.3 MHz, BW 220.8 MHz.
4. Run initial EM simulation with 2D mesh density of 20 cells/wavelength.
5. Optimize resonator geometry using gradient descent (target IL < 3.4 dB).
6. Verify coupling coefficients between resonator stages.
7. Run 3D FEM simulation to account for package parasitics (CSP (Chip Scale Package)).
8. Export GDS-II layout for mask fabrication.
9. Fabricate prototype wafers (6-inch, AlN on SiC).
10. Perform on-wafer probing using Rohde & Schwarz ZNA67.
11. Measure S-parameters (S11, S21, S12, S22) from 100 MHz to 16309 MHz.
12. Compare measured results against simulation and specification limits.
13. Perform temperature cycling (-40°C to +85°C) and re-measure.
14. Document results and generate summary report.

5. ACCEPTANCE CRITERIA

- Insertion loss at center frequency shall be <= 3.4 dB
- Return loss in passband shall be >= 18.0 dB
- Out-of-band rejection at +/- 331 MHz offset >= 46.0 dB
- Group delay variation across passband shall be <= 2 ns
- Temperature drift over -40°C to +85°C shall be <= 6.8 MHz
- Power handling shall be >= +31 dBm at 1 dB compression
- ESD tolerance >= 500 V (HBM)
- Die yield on qualification lot >= 88%

6. TEST RESULTS

Measurement Date: 2024-12-14
Devices Tested: 83
Insertion Loss (meas): 3.15 dB
Return Loss (meas): 20.2 dB
Rejection (meas): 48.6 dB
Isolation (meas): 32.0 dB
Q Factor (meas): 4117
Group Delay Variation: 1.5 ns
Power Handling: +28 dBm
Die Yield: 82.0%

Result Classification: FAIL

7. OBSERVATIONS AND FINDINGS

- a) The SAW (Surface Acoustic Wave) design demonstrated below-target performance for WiFi 5 GHz.
- b) Insertion loss is marginally above target; electrode thickness optimization recommended.
- c) Out-of-band rejection exceeded the minimum requirement.
- d) Temperature stability was within specification across full range.
- e) ESD robustness exceeded specification by 2x margin.

8. CORRECTIVE ACTIONS

- 1. Optimize resonator geometry to reduce insertion loss by ~0.3 dB.
- 2. Re-run EM simulation with refined mesh for better accuracy.
- 3. Escalate to advanced materials team for alternative piezoelectric stack.

9. SIGN-OFF

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